

To the ambiguity,

$$S \rightarrow A/B.$$

$$A \rightarrow i c t A e A / a.$$

$$B \rightarrow i c t S / i c t A e B.$$

$$C \rightarrow b.$$

29/9.

Monday.

Chomsky Hierarchy (5m ESE)

* The Chomsky suggested 4 different classes/phases.

1. Type 0 (Unrestricted Grammar)

2. Type 1 (Context Sensitive Grammar)

3. Type 2 (Context Free Grammar)

4. Type 3 (Regular Grammar).

* Identify the type of Grammar.

1. Type 0 (Unrestricted Grammar)

- There are no restrictions on the production of the grammar of this type.

- This type of grammar permits production of the form $\alpha \rightarrow \beta$ with $\alpha \neq \epsilon$.

where α, β are ^{sentential} ~~sentential~~ form, which is combination of terminals & non terminals.

So, this type of grammar is called as unrestricted grammar.

- Eg: Let $V = (A, B, C)$
 $T = (a, b, c)$

As we can see there is no restriction of productions.

$\therefore$ It is unrestricted grammar.

2. Type 1 (Context sensitive Grammar)

- The restrictions on this type of grammar are :

1. Each production of the form $\alpha \rightarrow \beta$, the length of β is as much as α , except $S \rightarrow \epsilon$.
2. The start symbol 'S' should not appear on right side of any production
3. The term 'Context sensitive' means :

$$\alpha_1 A \alpha_2 \rightarrow \alpha_1 \beta \alpha_2 \quad (\beta \neq \epsilon)$$

eg: Consider the grammar

$$G = (\{S\}, \{a, b\}, P, S) \quad G = (\{A, B, C\}, \{a, b\}, P, A)$$

$$S \rightarrow aSa | bsb | a | b$$

$$A \rightarrow BC$$

$$AB \rightarrow AC$$

$$C \rightarrow ab.$$

3. Type 2

- In this type of grammar, the allowed type of productions are : $\alpha \rightarrow \alpha$ $A \rightarrow \alpha$, where A is non-terminal, α is sentential form.

- eg : $G : (\{S\}, \{a, b\}, P, S)$

$$S \rightarrow aSa | bsb | a | b.$$

Sentential :

- non terminal-terminal.

4. Type 3

- There is a restriction on this type of grammar :

1. Left side of each production should contain only one non terminal.
2. Right side can contain atmost one non terminal which is allowed to appear as leftmost / rightmost symbol.

- Depending on position of Non terminal, the Grammar is divided into 2 types.

1. Left Linear Grammar. (LLG)
2. Right Linear Grammar. (RLG)

1. LLG

- In LLG, the allowed type of productions are :

$$A \rightarrow BW \xrightarrow{(\alpha)}$$

$$A \rightarrow \epsilon \text{ or } A \rightarrow W$$

where A, B are variables & W is a string of terminals.

- eg: Consider the Grammar $G = (\{S, B, C\}, \{a, b\}, P, S)$

$$S \rightarrow Ca / Bb.$$

$$C \rightarrow Bb.$$

$$B \rightarrow Ba / b.$$

2. RLG

- The production of the form :

$$A \rightarrow WB, A \rightarrow W \text{ (or) } A \rightarrow \epsilon.$$

where A, B are non terminals & W is a string of terminals.

- eg: Consider the Grammar $G = (\{S, A\}, \{0, 1, \epsilon\}, P, S)$

$$S \rightarrow 0A; A \rightarrow 10A / \epsilon.$$

1. Convert the following grammar to CNF.

$$S \rightarrow ABA$$

$$CNF \rightarrow AB | a.$$

$$A \rightarrow aB | \epsilon.$$

$$B \rightarrow bB | \epsilon. \quad (\rightarrow \text{remove } \epsilon \text{ \& then start})$$

$$\text{Let } R_1 \rightarrow a, R_2 \rightarrow b, R_3 \rightarrow BA.$$

$$S \rightarrow AR_3$$

* ϵ will not come in CNF.

$$A \rightarrow R_1 B | \epsilon$$

$$B \rightarrow R_2 B$$

2. Greibach NF:

Every CFG / CNF without ϵ can be generated by the grammar for which the production of the form $A \rightarrow a\alpha$ where A is variable, ' a ' is terminal and α is string of variables. This type of Grammar is called GNF.

In this normal form, restrictions on the productions are:

on RHS, should contain only one terminal symbol and that should be ^{the} leftmost ~~the~~ symbol on RHS, followed by 0 or more non terminals.

eg: Convert the following grammar to GNF:

$$S \rightarrow ABA | AB | BA | AA | A | B.$$

$$A \rightarrow aA | a$$

$$B \rightarrow bB | b$$

In GNF, the production is as follows:

$$A \rightarrow a\alpha$$

If The productions of S are not in GNF, to bring them in GNF, we can replace A 's and B 's by the production as:

$$A \rightarrow aA$$

Final Grammar in GNF is:

$$A \rightarrow a$$

$$S \rightarrow aBA | aB | bA | aA | a$$

$$B \rightarrow bB$$

$$B \rightarrow b$$

$$S \rightarrow aABA | aBA | aAB | aB | bBA | bA | aAA | aA$$

eg: Convert the following grammar

$$G = (\{A_1, A_2, A_3\}, \{a, b\}, P, A_1)$$

where P consists of

$$A_1 \rightarrow A_2 A_3$$

$$A_2 \rightarrow A_3 A_1 \mid b$$

$$A_3 \rightarrow A_1 A_2 \mid a.$$

Sol: $\rightarrow$ replace A_2 and A_3 by $A_2 \rightarrow b$ and $A_3 \rightarrow a$

$\therefore$ final grammar

$$A_1 \rightarrow b A_3$$

$$A_2 \rightarrow a A_1 \mid b$$

$$A_3 \rightarrow b A_3 A_2 \mid a.$$

* Pumping lemma -
self study.

* Applications of Grammar :

7/10/25
Tuesday

Pushdown Automata :

Equivalence of Left Linear to Right Linear Grammar

To perform the steps to convert RLG to its equivalent LLG.

Rules:

1. Represent the given RLG by a transition diagram with vertex labelled as $(V \cup \{E\})$ and transition labelled as $(T \cup \{E\})$
2. Interchange the position of initial & final states.
3. Reverse the direction of all Transitions Keeping the position of all intermediate state unchanged.
4. Rewrite the Grammar in Left Linear Fashion.

Q) $RLG \rightarrow LLG$.

- Convert the following grammar $RLG \rightarrow LLG$.

$$1) S \rightarrow bB$$

$$B \rightarrow bC$$

$$B \rightarrow aB$$

$$C \rightarrow a.$$

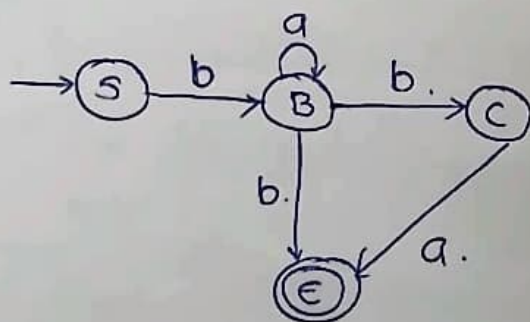
$$B \rightarrow b.$$

$$G = (\{S, B, C\}, \{a, b\}, P, S)$$

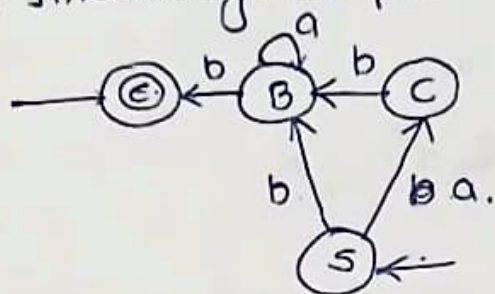
Sol:

Step 1: Draw the transition diagram and E as the final state.

S as the start symbol



Step 2: Interchange the position of initial & final states.



Step 3: In Reverse the direction of all Transitions

Step 4: Rewrite the Grammar in Left Linear Fashion.

$$S \rightarrow Ca.$$

$$S \rightarrow Bb.$$

$$C \rightarrow Bb$$

$$B \rightarrow Ba$$

$$B \rightarrow b.$$

Q) Derive the string "baab" using LLG & RLG.

Grammar: RLG:

$$S \Rightarrow bB \Rightarrow baB. \Rightarrow baab$$

LLG:

$$S \Rightarrow \cancel{b}B Bb \Rightarrow Bab. \Rightarrow Baab. \Rightarrow baab.$$

Q) Write an equivalent RLG to LLG:

$$S \rightarrow OA \mid IB.$$

$$A \rightarrow OC \mid IA \mid O.$$

$$B \rightarrow IB \mid IA \mid I.$$

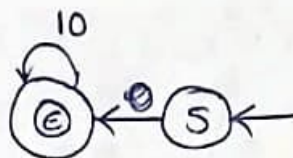
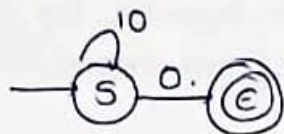
$$C \rightarrow O \mid OA.$$

Step 1: Draw the transition diagram based on Grammar.

Step 2: Interchange the position of initial and final state and reverse the direction

Step 3 Rewrite the grammar.

eg: $S \rightarrow S | 0 | 0.$



eg: $S \rightarrow B | A | C.$

$A \rightarrow C | A | B | 0.$

$B \rightarrow B | 1$

$C \rightarrow A | 0.$

eg: $S \rightarrow A.$

$A \rightarrow A | 0 | 0.$

29/9:

Step 1:

30/9:

Equivalence of RLG & FA

30/9 :

Equivalence of REG LLG & FA. (10m)

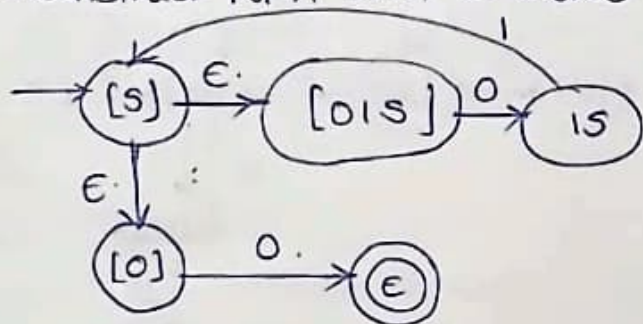
1. Find the equivalent DFA accepting the regular language defined by LLG given as follows:

$$S \rightarrow \overset{SIO}{\cancel{01S}} \mid 0$$

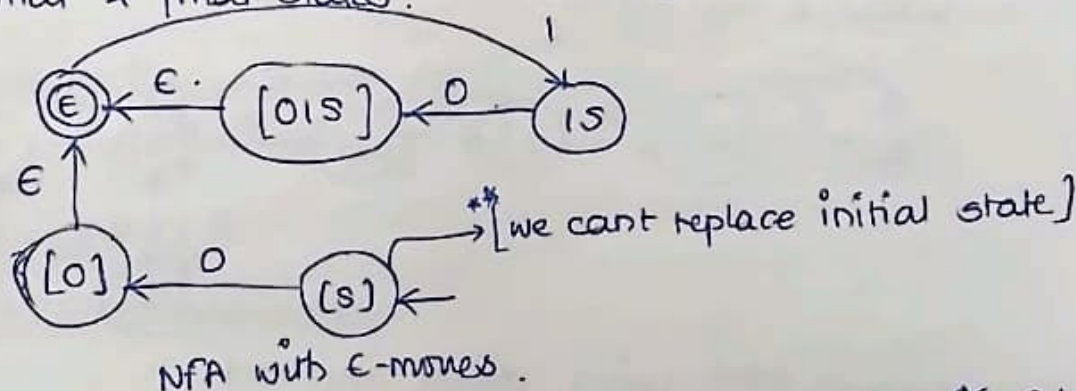
Step 1: Reverse the Right Hand side of all productions of LLG to obtain RLK

$$S \rightarrow 01S \mid 0$$

Step 2: Construct NFA with ϵ moves.

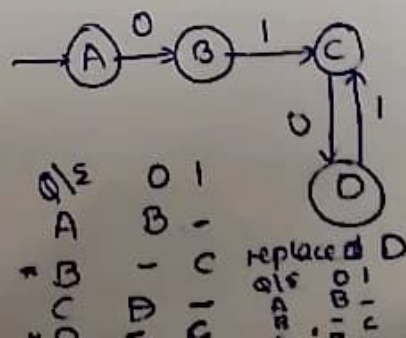
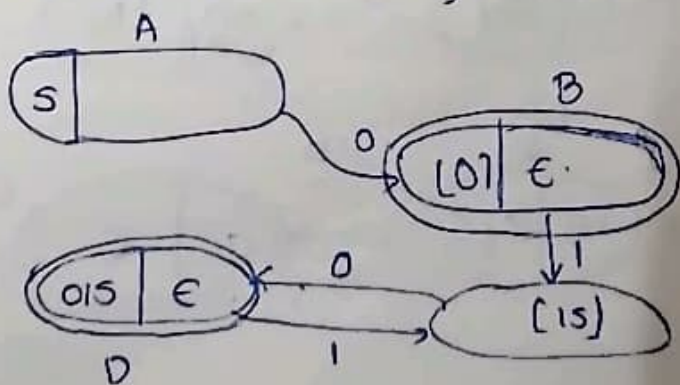


Step 3: Reverse the direction and interchange the position of initial & final states.



Step 4: Construct DFA using direct method. $\$$

0 1
A B
+ B - B



2. Construct DFA Accepting the regular language generated by

$$S \rightarrow Ca | Bb.$$

$$C \rightarrow Bb$$

$$B \rightarrow Ba | b.$$

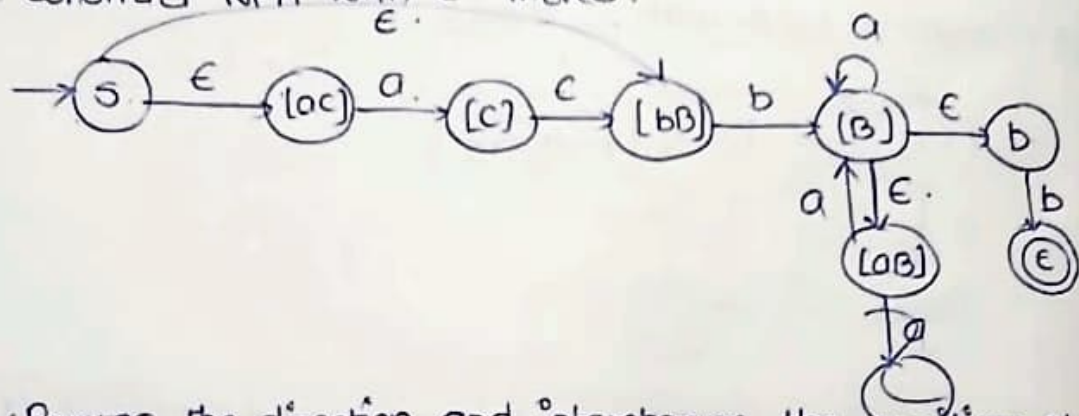
Step 1: Reverse the Right Hand Side of all productions of LLG to obtain RLG.

$$S \rightarrow aC | bB$$

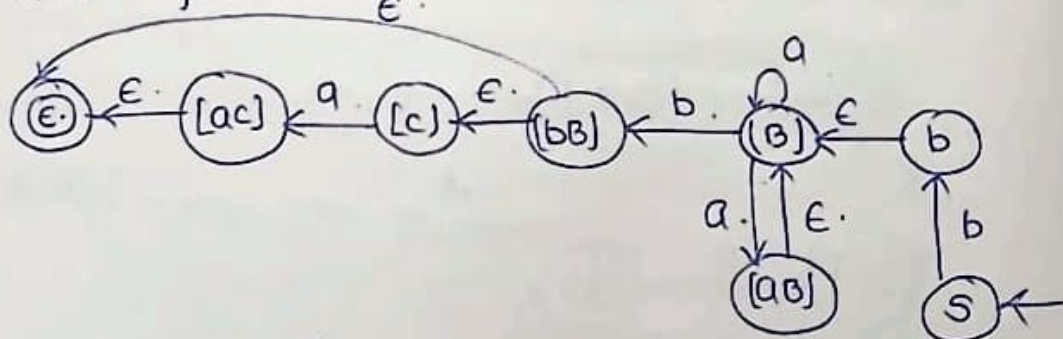
$$C \rightarrow bB.$$

$$B \rightarrow aB | b.$$

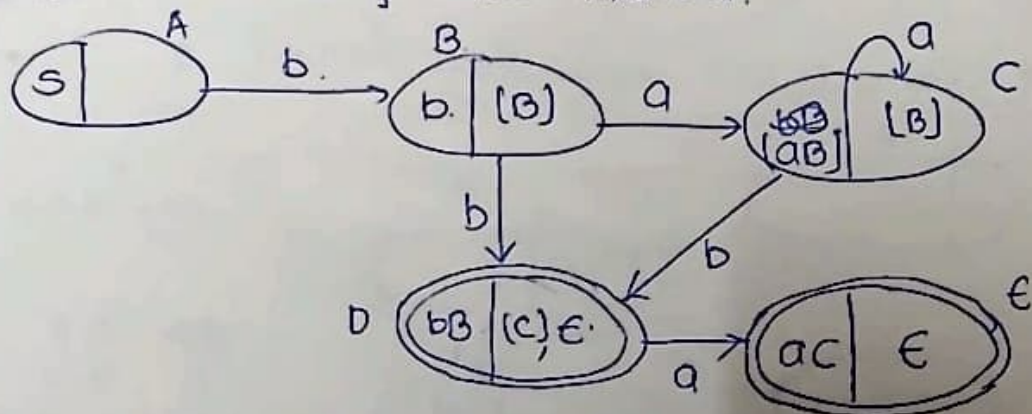
Step 2: Construct NFA with ϵ moves.



Step 3: Reverse the direction and interchange the position of initial & final states ϵ .



Step 4: Construct DFA using direct method.



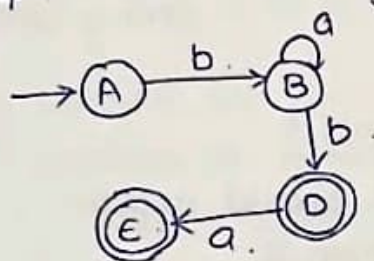
step 5: DFA diagram

Q \ Σ	a	b
A	-	B
B	C	D
C	C	D
* D	E	-
* E	-	-

step 6: DFA after reduction

Q \ Σ	a	b
A	-	B
B	B	D
* D	E	-
* E	-	-

step 7: Final DFA Diagram.



0110:

Simplification of CFG:

1. Removal of Useless Symbol.
2. Removal of unit production.
3. Elimination of ϵ -production.

$S \rightarrow aB | A$ $\therefore A$ is useless symbol here.
 $B \rightarrow b$ It is not deriving anything.

$S \rightarrow asa | bsb | \epsilon$

to remove ϵ -production:

$S \rightarrow asa$
 $S \rightarrow bsb$
 $S \rightarrow \epsilon$

$S \rightarrow asa$
 $S \rightarrow bsb$
 $S \rightarrow aa$
 $S \rightarrow bb$

removing S because $S \rightarrow \epsilon$.

$\overset{A}{\epsilon} \rightarrow$ Non terminal null string.

Simplification of CFG:

There are 3 ways to simplify CFG:

1. Removal of Useless symbol.
2. Removal of unit production.
3. Elimination of ϵ -production.

1. Removal of Useless symbol:

A Symbol α is useful if there exists a derivation

$$S \xRightarrow{*} \alpha \times \beta \xRightarrow{*} w, \text{ where } \alpha, \beta \text{ are sentential form \& } w \text{ is any string in } \downarrow \text{sentential form (VUT)}^*$$

There are 3 aspects of useless symbol A:

- 1) Some strings must be derived from A
- 2) A should appear at least once in the derivation derived from S.
- 3) It should not be non-terminal.

eg: $G = (\{S, A\}, \{1, 0\}, P, S)$

where P consists of:

$$S \rightarrow 10 \mid 0S1 \mid 1S0 \mid A \mid SS$$

A is useless symbol

As we can see $S \Rightarrow A \Rightarrow ?$ this means A is Useless symbol.

Removing the useless symbol means deleting the entry of A
 $\therefore$ The final grammar is represented as:

$$S \rightarrow 10 \mid 0S1 \mid 1S0 \mid SS.$$

eg2: $G = (\{S, A, B\}, \{0\}, P, S)$

$$S \rightarrow AB \mid a$$

$$A \rightarrow a$$

$\therefore$ Here, B is useless symbol here, since it is not deriving anything.

2. Removal of Unit Production.

A Production of the form $A \rightarrow B$, where $A \neq B$ are non terminals. Such productions are called as unit production

Elimination Rules:

- For every pair of production, if non terminals in the form of $A \neq B$

Rule 1:

If the CFG has unit production $A \rightarrow B$ then there is a chain/cycle. $A \Rightarrow X_1 \Rightarrow X_2 \dots \Rightarrow X_n$. (to solve unit productions, we make a chain.)

eg: Consider the Grammar $G = (\{A, B\}, \{a, b\}, P, A)$ where P consists of $A \rightarrow B$
 $B \rightarrow a | b$. // unit productions

Sol: $A \Rightarrow B \Rightarrow a | b$.

$A \rightarrow a | b$.

eg2: Consider the Grammar:
$$\left. \begin{array}{l} S \rightarrow A | bb \\ A \rightarrow B | b \\ B \rightarrow S | b \end{array} \right\} \text{Sol: } S \rightarrow A \rightarrow B \rightarrow S.$$

eg3: $S \rightarrow A | bb$. $\text{sol: } a | b | bb$. $S \rightarrow a | b | bb$.
 $A \rightarrow S | a | b$.

3. Elimination of ϵ -production

The production of the form $A \rightarrow \epsilon$, where A is non terminal also called as nullable non terminal
(value should be 0)

eg: Consider the Grammar

$$\left. \begin{array}{l} A \rightarrow B | \epsilon \\ B \rightarrow A \end{array} \right\} \Rightarrow \begin{array}{l} A \rightarrow B \\ A \rightarrow \epsilon \\ B \rightarrow A \end{array}$$

Sol: In this Grammar, A is nullable, non terminal B is not nullable.

$\therefore$ Simplified Grammar:

$A \rightarrow B$

$B \rightarrow a$.

eg2: $A \rightarrow B|a$.

$B \rightarrow A|\epsilon$.

As, $B \Rightarrow \epsilon$, so B is nullable.

There is a chain $A \Rightarrow B \Rightarrow \epsilon$ Sol: $A \rightarrow a$ ✓

$\therefore$ Simplified form: $A \xRightarrow{*} \epsilon$.

$\therefore A$ is also nullable.

Nullable Non-terminal — where ϵ is present.

Elimination Procedure: / Rules :

1. Delete all ϵ productions from the grammar.
2. Identify Nullable Non Terminal.
3. If there is a production of the form, $A \rightarrow \alpha$, where A is the sentential form consist of atleast one nullable non terminal, Then add new productions and delete all possible nullable non terminal entries.
4. If using step 3, if we get the production of the form $A \rightarrow \alpha$ then dont add that entries in the final state / final Grammar.

6/09

Monday.

1. Eliminate ϵ -production from G , where G is represented as:
(2m) $S \rightarrow asa|bsb|\epsilon$.

Sol: $S \rightarrow asa$

Step 1: Delete the ϵ -production.

$S \rightarrow asa$.

$S \rightarrow bsb$. (or) $S \rightarrow asa|bsb$

Step 2: Finding the nullable non-terminal
as $S \rightarrow \epsilon$ so S is nullable non-terminal

Step 3: Adding new productions.

$S \rightarrow asa|bsb|aa|bb$ (Ans)

There are 2 productions consisting of nullable non terminals

$S \rightarrow asa$

$S \rightarrow bsb$

If we delete $S \rightarrow aSa$ from that S ; if we delete S from RHS, we can add new production as $S \rightarrow aa$.

For $S \rightarrow bSb$, if we delete S from right hand side, we can add new production as $S \rightarrow bb$.

$\therefore$ The final grammar is represented as: $S \rightarrow aSa | bSb | aa | bb$
 $L(G') = L(G) - L(\epsilon)$.

2. Eliminate ϵ -production from Grammar G given as:

$$S \rightarrow a | xb | aYa.$$

$$X \rightarrow Y | \epsilon.$$

$$Y \rightarrow b | X$$

Sol: $\Rightarrow$

Step 1: Delete the ϵ -production.

$$S \rightarrow a \quad X \rightarrow Y \quad Y \rightarrow b. \quad \dots$$

$$S \rightarrow xb \quad X \rightarrow \epsilon \quad Y \rightarrow X$$

$$S \rightarrow aYa. \quad \hookrightarrow X \text{ is nullable non terminal.}$$

$$Y \Rightarrow X \Rightarrow \epsilon.$$

$$Y \stackrel{*}{\Rightarrow} \epsilon. \rightarrow Y \text{ is also nullable non terminal.}$$

$$\begin{matrix} S \rightarrow a & S \rightarrow a | xb | aYa. & \therefore X \rightarrow Y. \\ S & \end{matrix}$$

Step 2: Finding the nullable Non terminal.

X, Y are nullable non terminals.

Step 3: Adding new productions

$$S \rightarrow a | xb | aYa. | aa. | b. \quad \text{remove the nullable non terminal and write}$$

$$X \rightarrow Y.$$

$$Y \rightarrow b | X.$$

3. Consider the Grammar
- $$S \rightarrow ABA$$
- $$A \rightarrow aA \mid \epsilon.$$
- $$B \rightarrow bB \mid \epsilon.$$

Normal Forms :

* (Compulsory)

1. There are certain restrictions while writing ~~restrict~~ the grammar
— are called as Normal forms.

There are 2 types of Normal forms:

1. Chomsky NF. (CNF)
 2. Greibach NF (GNF)
- } * will not have any ϵ -transition

1. Chomsky NF.

→ Any context free language without ϵ -production is generated by a grammar in which all the production of the form

$A \rightarrow BC$ (or) $A \rightarrow a$. // 2 non terminals (or) 1 terminal must be present on RHS
where A, B, C are the variables and ' a ' is a terminal is called as CNF.

eg: Convert the following CFG to CNF.

$S \rightarrow aSa \mid bSb \mid a \mid b \mid aa \mid bb$ // Move to be in the form:
 $A \rightarrow BC \mid A \rightarrow a$

IN CNF, we have two types of productions:

$A \rightarrow BC$

$A \rightarrow a.$

If we add 2 productions namely:

$S \rightarrow a \quad A \rightarrow a$

$S \rightarrow b \quad B \rightarrow b$

We can write the above grammar as:

$S \rightarrow ASA \mid BSB \mid a \mid b \mid AA \mid BB.$

We need not change the production of the form $S \rightarrow a$ and $S \rightarrow b$ because they are already in CNF.

Now there are only 2 productions :

$S \rightarrow ASA$ and $S \rightarrow BSB$, which are not in CNF.

To introduce the new variable R_1 & R_2 such that

$R_1 \rightarrow SA$, $R_2 \rightarrow SB$.

$\therefore$ CNF :

$S \rightarrow AR_1 \mid BR_2 \mid a \mid b \mid AA \mid BB$

$A \rightarrow a$

$B \rightarrow b$

$R_1 \rightarrow SA$

$R_2 \rightarrow SB$.

eg: Convert the following grammar using CNF.

$S \rightarrow bA \mid aB$.

$A \rightarrow bAA \mid aS \mid a$

$B \rightarrow aBB \mid bS \mid b$.

In CNF, we have two types of productions :

$A \rightarrow BC$

$A \rightarrow a$

If we add 2 productions namely :

$R_1 \rightarrow a$

$R_2 \rightarrow b$

$S \rightarrow R_1A \mid R_2B$

$A \rightarrow R_2AA \mid R_1S \mid a$.

$B \rightarrow R_1BB \mid R_2S \mid b$.

we need not change the production of the form $A \rightarrow a$ and $B \rightarrow b$ because they are already in CNF.

Now there are only 2 productions : $A \rightarrow R_2AA$ & $B \rightarrow R_1BB$ which are not in CNF.

To introduce new variable R_3 & R_4 such that :

$R_3 \rightarrow AA$ and $R_4 \rightarrow BB$.

$\therefore$ CNF : $S \rightarrow R_1A \mid R_2B$

$A \rightarrow R_2R_3 \mid R_1S \mid a$

$B \rightarrow R_1R_4 \mid R_2S \mid b$